

Exact maximal order types of finite-alphabet transfinite words

A separator construction for all lengths below ω^k

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Abstract

For a finite poset P , let $\mathcal{W}_k(P)$ be the quasi-order of P -labelled words of ordinal length less than ω^k , ordered by subsequence embedding. Let $N_k(P)$ be the number of equivalence classes of nonempty indecomposable words in this range. We give a proof of the formula

$$o(\mathcal{W}_k(P)) = \omega^{\omega^{N_k(P)-1}} \quad (|P| \geq 2, 1 \leq k < \omega).$$

The number $N_k(P)$ is finite and effectively computable from a canonical poset of bounded-height periodic terms. In particular, if $J(P)$ counts all order ideals, including the empty ideal, then

$$o(\mathcal{W}_2(P)) = \omega^{\omega^{|P|+J(P)-2}}.$$

For the two-element antichain this is ω^{ω^4} , the upper bound whose tightness is left unverified in Altman's Example 3.15. For the two-element chain it is ω^{ω^3} . The lower-bound argument does not assume that concatenation preserves incomparability: it uses ordinal-length protection to prevent absorption at separators. The package includes full proofs, exact atom-poset computations, and reproducible symbolic embedding tests.

Research status. This is a proposed solution with a complete written proof, not an independently refereed or machine-formalized result. The source problem was checked against arXiv:2409.03199v2, the version listed on September 19, 2026. The bounded-height tree/indecomposable correspondence is established background; the claimed contribution here is attainment of the finite-generator upper bound by protected separator constructions.

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1 The problem and the claimed answer

The starting point is a very concrete question about two letters. Write $A_2 = \{a, b\}$ with a and b incomparable. A word is allowed to have any ordinal length less than ω^2 : a finite number of infinite blocks, possibly followed by a finite suffix. One word embeds in another when its positions can be mapped increasingly to positions carrying the same letters. How large can the order type of a well-ordered linear extension of this embedding order be, after mutually embeddable words are identified?

Altman gives the upper bound ω^{ω^4} and explicitly leaves its attainment unverified in Example 3.15 of [1]. In the same example, he observes that the two-element chain has an improved upper bound ω^{ω^3} , because two of the periodic building blocks coincide up to embedding equivalence. The arXiv record lists version 2, revised March 10, 2026; the example is on page 10 of that version. These are the specific finite-alphabet questions addressed here.

There are two reasons to attack the binary case through a more general statement. First, the mechanism responsible for an additional ordinal exponent is not specific to either binary alphabet. Second, once the correct lower-bound construction is found, the same construction can be iterated at every finite ordinal level.

1.1 Main theorem

Precise definitions follow in sections 2 and 3. For now, $\mathcal{I}_k(P)$ denotes the poset of equivalence classes of nonempty indecomposable P -words of lengths

$$1, \omega, \omega^2, \dots, \omega^{k-1},$$

and $N_k(P) = |\mathcal{I}_k(P)|$. A word is *indecomposable* here when it is mutually embeddable with every nonempty final segment of itself.

Theorem 1.1 (Finite-alphabet exactness). *Let P be a finite poset with $n \geq 2$ elements, and let $k \geq 1$ be finite. Then $\mathcal{I}_k(P)$ is finite and*

$$\mathfrak{o}(\mathcal{W}_k(P)) = \omega^{\omega^{N_k(P)-1}}. \quad (1)$$

Writing $J(R)$ for the number of order ideals of a finite poset R , including the empty ideal, the numbers are computed by

$$\mathcal{I}_0(P) = \emptyset, \quad N_1(P) = n, \quad N_{k+1}(P) = n + J(\mathcal{I}_k(P)) - 1. \quad (2)$$

The recurrence requires the actual order on $\mathcal{I}_k(P)$, not just its cardinality.

The full proof is in section 7. Two consequences can already be stated.

Corollary 1.2 (The first transfinite level). *If P is a finite poset of size $n \geq 2$, then*

$$\mathfrak{o}(\mathcal{W}_2(P)) = \omega^{\omega^{n+J(P)-2}}. \quad (3)$$

In particular,

$$\mathfrak{o}(\mathcal{W}_2(A_2)) = \omega^{\omega^4}, \quad \mathfrak{o}(\mathcal{W}_2(C_2)) = \omega^{\omega^3},$$

where C_2 is the two-element chain.

Corollary 1.3 (Necessary exceptional cases). *For the empty alphabet, $\text{o}(\mathcal{W}_k(\emptyset)) = 1$. For the one-element alphabet,*

$$\text{o}(\mathcal{W}_k(1)) = \omega^k.$$

A finite quasi-order is first replaced by its quotient under mutual comparison; the number of classes, not the number of original labels, determines whether these exceptions apply.

Proof. There is only the empty word over the empty alphabet. Over one letter, a word is determined by its length up to equivalence, and embedding is exactly ordinal comparison of lengths. The set of lengths less than ω^k has order type ω^k . $\square$

1.2 What is established background, and what is being added?

The maximal-order-type calculus for well-quasi-orders is due to de Jongh and Parikh and to subsequent work, particularly Schmidt [3, 5]. Higman’s finite-word construction and its exact finite-alphabet order type are not new here [4, 5]. Chopra and Pakhomov establish a correspondence between finite leaf-labelled trees and indecomposable finite-image sequences below ω^ω [2, Theorem 4.26]. The normal forms used below are a bounded-height, antichain-reduced presentation of that existing structure.

The point requiring a new argument is the lower bound in [theorem 1.1](#). A finite list of indecomposable types gives an immediate Higman upper bound. It does *not* give a matching lower bound by simply concatenating those types: for example,

$$a a^\omega \equiv a^\omega.$$

The proof below constructs large induced suborders on which this absorption is prevented by a final segment of sufficient ordinal length. Every new indecomposable type can then serve as a separator and contribute one more step in the finite exponent parameter.

The claim of novelty is limited to that exactness argument and its consequences. The sources consulted did not supply this finite-level attainment theorem; this is not an exhaustive determination of priority. The theorem for finite alphabets is also not a claim to settle every infinite-alphabet lower-bound question.

2 Words, quotients, and ordinal accounting

2.1 Transfinite words and embedding

Definition 2.1. A P -word of length α is a function $u: \alpha \rightarrow P$. For words $u: \alpha \rightarrow P$ and $v: \beta \rightarrow P$, write $u \preccurlyeq v$ if there is a strictly increasing map $f: \alpha \rightarrow \beta$ such that $u(\xi) \leq_P v(f(\xi))$ for every $\xi < \alpha$. Set $u \equiv v$ when both $u \preccurlyeq v$ and $v \preccurlyeq u$.

Concatenation is indexed by ordinal addition: if $|u| = \alpha$ and $|v| = \beta$, then $|uv| = \alpha + \beta$. Finite and countable concatenations preserve embedding in every component. In particular, replacing the components of a concatenation by equivalent words produces an equivalent word.

For a positive integer k ,

$$\mathcal{W}_k(P) = \{u : |u| < \omega^k\},$$

with the empty word included. Since P is finite, every such word has finite image. This is the object denoted $s_{\omega^k}^F(P)$ in the source problem. All maximal order types below refer to the quotient by $\equiv$.

An increasing map between ordinals can exist only from a smaller ordinal to a larger one. Consequently

$$u \preceq v \implies |u| \leq |v|, \quad u \equiv v \implies |u| = |v|. \quad (4)$$

This elementary fact makes word length a well-defined invariant of an equivalence class. It is ordinal length, not cardinality, that is relevant: a bounded interval in ω^2 may be countably infinite while still being too short to contain an increasing copy of ω^2 .

2.2 Maximal order type

A quasi-order is a *well-quasi-order* (wqo) if every infinite sequence $x_0, x_1, \dots$ has $i < j$ with $x_i \leq x_j$. Its maximal order type $\text{o}(X)$ is the largest order type of a linear extension of its quotient poset. The existence of a largest type, rather than merely a supremum, is part of the de Jongh–Parikh theory [3]. We use the following standard facts:

$$\begin{aligned} \text{o}(X \sqcup Y) &= \text{o}(X) \oplus \text{o}(Y), \\ \text{o}(X \times Y) &= \text{o}(X) \otimes \text{o}(Y), \\ \text{o}(X) &= \sup_{x \in X} (\text{o}(X \setminus \uparrow x) + 1). \end{aligned} \quad (5)$$

Here $\sqcup$ is the disjoint union, the product is componentwise, and $\oplus, \otimes$ are natural ordinal addition and multiplication. An induced suborder or, more generally, the domain of an order-reflecting map into a wqo has no larger maximal order type. A monotone surjection from a wqo onto another quasi-order, also allowing surjectivity up to equivalence, gives the reverse comparison of domain and image types. These are standard tools in the same calculus [3, 5].

For an integer $m \geq 1$ put

$$H_m = \omega^{\omega^{m-1}}. \quad (6)$$

The finite-alphabet Higman formula is

$$\text{o}(R^*) = H_{|R|} \quad (0 < |R| < \omega), \quad (7)$$

where R^* is the finite-word order over the finite poset R . For completeness, [section A](#) derives the finite-alphabet formula from the standard wqo calculus. The order relations among the letters do not change (7); they will matter for the transfinite atom count.

2.3 The ordinal amplification identity

The numerical calculation used repeatedly below is simple but important. For $m, d \geq 1$,

$$H_m^{\otimes d} = \omega^{\omega^{m-1} \cdot d}. \quad (8)$$

Indeed, multiplying monomials naturally adds their exponents naturally, and the natural sum of d copies of ω^{m-1} is $\omega^{m-1} \cdot d$. The countable *ordinary* ordinal sum satisfies

$$\sum_{d=1}^{\infty} H_m^{\otimes d} = \omega^{\omega^{m-1} \cdot \omega} = \omega^{\omega^m} = H_{m+1}. \quad (9)$$

The same expression is obtained by indexing $d = m' + 1$ for $m' \geq 0$. No countable natural product is being invoked.

Lemma 2.2 (Layered products). *Suppose a wqo Y contains induced copies Y_d of E^d for all $d \geq 1$, where $\text{o}(E) = H_m$. Suppose the copies are pairwise disjoint on the quotient and*

$$x \in Y_d, y \in Y_e, x \leq y \implies d \leq e.$$

Then $\text{o}(Y) \geq H_{m+1}$.

Proof. For each d , choose a maximal linear extension of Y_d . Order the layers by increasing d and use the chosen extension within each layer. The hypothesis guarantees that this is a linear extension of their *induced union*. Its order type is the left side of (9), by the product rule. The induced union is a suborder of Y , so its type gives a lower bound for $\text{o}(Y)$. There is no assertion that this particular linear extension must extend to all of Y with the same type. $\square$

3 Canonical indecomposables at finite height

This section supplies a self-contained bounded-height normal form. It is included to make the integer $N_k(P)$ effective and to identify precisely which generators are used by the lower-bound proof. The general tree/sequence correspondence is established in [2].

3.1 Indecomposability and finite concatenations

For a word $u: \alpha \rightarrow P$ and $\xi < \alpha$, its tail at ξ is the restriction to $[\xi, \alpha)$, reindexed by that interval's order type.

Definition 3.1. A nonempty word u is indecomposable if it is equivalent to every one of its nonempty tails.

If $|u| < \omega^k$ and u is indecomposable, then $|u| = \omega^r$ for some $r < k$. To see this, write $|u|$ in Cantor normal form. Unless it is a single power of ω with coefficient one, it has a nonempty tail of smaller order type. That contradicts (4). The property is invariant under equivalence: an embedding between two words of the same indecomposable ordinal length is cofinal, so a tail in one word receives a tail of the other.

Lemma 3.2 (An indecomposable lands in one factor). *If z is indecomposable and*

$$z \preceq u_1 u_2 \cdots u_t$$

for a finite list of words, then $z \preceq u_i$ for some i .

Proof. Among the finitely many target factors met by the embedding, choose the last one. From any source position whose image lies in that factor onward, all source positions map into the same factor. Thus a nonempty tail of z embeds in it. That tail is equivalent to z . $\square$

The lemma does not say that the original embedding places the entire source inside one factor. It says that *some* embedding does so. This distinction will also matter for separators.

3.2 Periodic terms and their order

Start with a leaf p of height zero for each $p \in P$. Given a finite nonempty set S of previously constructed terms, introduce

$$\Omega(S), \quad \text{ht}(\Omega(S)) = 1 + \max_{s \in S} \text{ht}(s).$$

We will retain only sets S that are antichains under the already defined term order. Choose any enumeration $S = \{s_1, \dots, s_t\}$ and define

$$\text{ev}(p) = p, \quad \text{ev}(\Omega(S)) = (\text{ev}(s_1) \cdots \text{ev}(s_t))^\omega. \quad (10)$$

The particular enumeration changes neither the embedding class nor the comparison rule below. Every evaluation is indecomposable, and

$$|\text{ev}(s)| = \omega^{\text{ht}(s)}. \quad (11)$$

For a node, every tail contains all sufficiently late complete periods, which proves indecomposability. The length calculation follows by taking the largest height among the finitely many factors of a period and then multiplying the period length by ω on the right.

Proposition 3.3 (Comparison of periodic terms). *The following recursive rules agree exactly with semantic embedding:*

$$p \leq q \iff p \leq_p q, \quad (12)$$

$$p \leq \Omega(T) \iff (\exists t \in T) p \leq t, \quad (13)$$

$$\Omega(S) \leq p \text{ never}, \quad (14)$$

$$\Omega(S) \leq \Omega(T) \iff (\forall s \in S)(\exists t \in T) s \leq t. \quad (15)$$

In particular, comparison terminates by descending to lower-height terms.

Proof. The leaf rules follow directly from the presence or absence of a suitable letter. A positive-height word cannot embed into a word of length one.

For necessity in (15), fix one occurrence of a source factor $\text{ev}(s)$ inside a period of $\text{ev}(\Omega(S))$. There is a later source position, so the image of this factor is bounded in the target word. A bounded initial segment of a countable repetition of a finite period meets only finitely many complete or partial periods. Therefore $\text{ev}(s)$ embeds into a finite concatenation of target factors $\text{ev}(t)$. By lemma 3.2, it embeds into one of them. This holds for each $s \in S$.

Conversely, choose for each s a suitable t with $s \leq t$. Place successive source factors into suitably chosen target factors in successive fresh target periods. Repeating this construction through ω source periods gives an embedding. The argument with $S = T$ but different enumerations proves independence of the enumeration. $\square$

For a finite set S of terms, remove every element strictly below another element of S , retaining the maximal elements. This changes neither side of the Hoare comparison in (15). Thus every node has an antichain-reduced form. Moreover, if two finite antichains dominate one another, then they are equal: for $s \in S$, choose $t \in T$ and $s' \in S$ with $s \leq t \leq s'$; the antichain property forces $s = s'$, and antisymmetry forces $s = t$.

Induction on height now proves that the reduced terms form a genuine poset, with no distinct but equivalent terms. Leaves and nodes cannot be equivalent because their lengths differ.

3.3 Completeness and finiteness

Theorem 3.4 (Finite decomposition and canonical atoms). *For every finite P and integer $k \geq 1$:*

1. *there are finitely many reduced terms of height less than k ;*
2. *every P -word of length less than ω^k is equivalent to a finite concatenation of evaluations of these terms;*
3. *every indecomposable word of length ω^r , $r < k$, is equivalent to exactly one reduced term of height r .*

Consequently those terms, with the order in proposition 3.3, are precisely $\mathcal{I}_k(P)$.

Proof. For $k = 1$ the words are finite and the only indecomposables are letters. Assume the result through height $r - 1$ and consider a word u of length ω^r , where $r \geq 1$.

Partition its domain into ω consecutive intervals of length ω^{r-1} . By induction, each resulting block is equivalent to a finite nonempty list of lower-height atoms. Replacing all blocks by those lists and flattening yields an ω -indexed list of atoms of heights at most $r - 1$. There are only finitely many available atom types.

Let S be the nonempty set of types occurring infinitely often in that list. Delete a finite initial part containing all occurrences of every type outside S . The remaining list is equivalent to a cyclic repetition of S : map each of its successive entries into a fresh target period in one direction, and choose successive sufficiently late occurrences of the required entries in the other. Replacing S by its maximal elements gives a reduced node $\Omega(S)$.

Every original block has length ω^{r-1} , so its finite factorization must contain an atom of height $r - 1$. Finitely many types are available; hence some such type recurs infinitely often. Therefore the reduced set S still has maximal height $r - 1$, and its node has height r . We have shown that u is equivalent to a finite lower-height prefix followed by one atom of height r .

If u itself is indecomposable, it is equivalent to that last atom. One way to justify this without assuming a particular factorization of u is to use lemma 3.2: the finite prefix factors have length less than ω^r , so u cannot embed into them. It must embed into the final atom, which already embeds into u .

For an arbitrary ordinal $\alpha < \omega^{r+1}$, split α into finitely many blocks of length ω^r and one remainder of smaller length. The preceding argument handles the blocks and induction handles the remainder. This proves finite decomposition at the next level. There are only finitely many nonempty antichains of the finite poset of lower-height terms, so there are finitely many new terms. Uniqueness and exact comparison follow from proposition 3.3 and antichain reduction. $\square$

3.4 The exact atom-count recurrence

Every order ideal of a finite poset is determined by its antichain of maximal elements, and every antichain determines an ideal. Empty objects correspond to one another. Thus $J(R)$ also counts the antichains of R .

All nonleaf terms of height at most k are in bijection with the nonempty antichains of $\mathcal{I}_k(P)$. This includes lower-height nodes already present: a node with lower-height children is the same node, not a new copy. Adding the n leaves gives

$$N_{k+1}(P) = n + J(\mathcal{I}_k(P)) - 1. \quad (16)$$

Equivalently, the number of *new* height- k atoms is $J(\mathcal{I}_k(P)) - J(\mathcal{I}_{k-1}(P))$. This formulation avoids double-counting old nodes.

Corollary 3.5 (The finite-generator upper bound). *If P is nonempty, then*

$$o(\mathcal{W}_k(P)) \leq H_{N_k(P)}.$$

More generally, if B is any finite nonempty set of distinct indecomposable word classes and $\mathcal{L}(B)$ is its finite-concatenation language, then $o(\mathcal{L}(B)) \leq H_{|B|}$.

Proof. Concatenation gives a monotone map from B^* onto $\mathcal{L}(B)$ up to equivalence. Apply the Higman formula and the monotone-surjection rule. For $B = \mathcal{I}_k(P)$, [theorem 3.4](#) identifies the image with $\mathcal{W}_k(P)$. The same argument proves that all these languages are wqo. $\square$

4 Cofinal blocks and protected final segments

Fix throughout this section

$$\lambda = \omega^r, \quad 1 \leq r < \omega.$$

Every nonempty final interval of λ has order type λ . Every bounded initial interval of λ has order type less than λ . It follows that an increasing embedding of a word of length λ into another word of that length has cofinal image.

4.1 The equal-block principle

Lemma 4.1 (Equal blocks force coordinate preservation). *Let $u_1, \dots, u_d$ and $v_1, \dots, v_d$ be words, all of length exactly λ . Then*

$$u_1 \cdots u_d \preceq v_1 \cdots v_d \iff u_i \preceq v_i \text{ for every } i. \quad (17)$$

More generally, an embedding from d such blocks into e such blocks requires $d \leq e$.

Proof. For each source block, consider the last target block it meets. A tail of the source block lies entirely in that target block. Its length is still λ , so its image is cofinal there. Successive source blocks must therefore have strictly increasing last target blocks: a later source position cannot be placed after a cofinal image while remaining in the same target block. This proves $d \leq e$.

When $d = e$, the strictly increasing assignment from source block indices to their last target block indices must be the identity. The first source block is contained in the first target block. After its cofinal image, no position of the second source block can lie in the first target block; and its last target block is the second. Continue inductively. Thus each source block lies wholly in its corresponding target block, proving necessity in (17). Sufficiency is componentwise concatenation of embeddings. $\square$

Indecomposability of the individual words is not assumed in this lemma. Their common ordinal length supplies the necessary cofinality. Equal numbers of blocks, however, are essential to its coordinate conclusion.

4.2 The protection condition

Definition 4.2. A word is λ -ended if its length is λd for some positive finite integer d .

Every nonempty tail of a λ -ended word has order type at least λ . More precisely, its length is λe for some positive finite e . A tail may contain several remaining λ -blocks; it need not have length exactly λ .

Lemma 4.3 (No bounded spill into a protected block). *Let u be λ -ended, and let z have length λ . In an embedding, a nonempty tail of u cannot be mapped into a bounded initial interval of a target occurrence of z .*

Proof. Such a tail has length at least λ , whereas the target interval has length less than λ . This violates (4). $\square$

This tiny observation is the protection mechanism. Without it, an old finite suffix can disappear into the beginning of a new infinite separator, destroying coordinate reflection.

5 Activating the first atom at a new level

The first atom of a new height needs a different construction from the later atoms. At this point there is not yet a large family of λ -ended words to place between separators.

Choose a minimal letter $a \in P$ and another letter $b \neq a$. Then $b \not\leq_P a$. Put

$$c = a^\lambda.$$

This is an indecomposable of length λ . It is minimal among the indecomposables of that length: any word of length λ embedding into it can use only letters at most a , hence only the letter a .

Lemma 5.1 (A terminal marker protects the whole lower-level word). *The map*

$$F: \mathcal{W}_r(P) \longrightarrow \{\text{words of length } \lambda\}, \quad F(u) = u b a^\lambda \quad (18)$$

is an order embedding on equivalence classes. Its image is contained in $\mathcal{L}(\mathcal{I}_r(P) \cup \{c\})$.

Proof. Since $|u| < \lambda$ and λ is an infinite additively indecomposable ordinal, $|u| + 1 + \lambda = \lambda$. Concatenation proves monotonicity.

Suppose $u b a^\lambda \preccurlyeq v b a^\lambda$. The distinguished source occurrence of b cannot map into the target a^λ tail, because $b \not\leq_P a$. It must map either to the distinguished target b or into v . Every position of u maps strictly before that position, and therefore inside v . Hence $u \preccurlyeq v$.

Finally, u is a finite concatenation of lower-height atoms up to equivalence, b itself is a lower-height atom, and $a^\lambda = c$. $\square$

It is not necessary that the distinguished source b map to the *distinguished* target b . The proof deliberately allows it to land earlier inside v ; either possibility recovers $u \preccurlyeq v$.

Proposition 5.2 (First-atom amplification). *Suppose $\text{o}(\mathcal{W}_r(P)) = H_N$, where $N = N_r(P)$. Then the language $\mathcal{L}(\mathcal{I}_r(P) \cup \{a^\lambda\})$ contains a λ -ended induced suborder of maximal order type H_{N+1} . The language itself has the same maximal order type.*

Proof. Let D be the image of F . By lemma 5.1, $\text{o}(D) = H_N$ and every word in D has length λ . For every $d \geq 1$, concatenate d independently chosen elements of D . By lemma 4.1, the resulting induced suborder is a copy of D^d .

The images for different d have distinct lengths λd , so their quotient classes are disjoint. Length comparison prohibits an embedding from a higher- d layer to a lower- d layer. Their union is λ -ended, and lemma 2.2 gives type at least H_{N+1} . There are exactly $N + 1$ available indecomposable generators, so corollary 3.5 gives the matching upper bound for the whole language and hence for this induced suborder. $\square$

This is the only point in the main induction where the availability of a second inequivalent letter is required. For a singleton alphabet no such b exists, and the much smaller order type in corollary 1.3 is the actual answer.

6 Separator amplification

We now prove the main lower-bound lemma. It permits new generators that dominate old ones, so it is not merely a construction with pairwise incomparable separators.

Lemma 6.1 (Protected separators). *Let B be a finite set of indecomposable word classes, each of length at most λ . Let z be an indecomposable word of length λ such that*

$$z \not\leq b \quad \text{for every } b \in B. \quad (19)$$

Let $E \subseteq \mathcal{L}(B)$ be an induced suborder consisting of λ -ended words. For $m \geq 0$, define

$$\Phi_m(u_0, \dots, u_m) = u_0 z u_1 z \cdots z u_m, \quad u_i \in E. \quad (20)$$

Then Φ_m is an order embedding of E^{m+1} into $\mathcal{L}(B \cup \{z\})$. Moreover,

$$\Phi_m(\vec{u}) \preceq \Phi_t(\vec{v}) \implies m \leq t. \quad (21)$$

All outputs are λ -ended.

Proof. First, z does not embed into any word of $\mathcal{L}(B)$. Indeed, otherwise lemma 3.2 would embed it into one generator, contrary to (19).

Step 1: each source separator has a target separator as its last component. Consider an embedding of an output of Φ_m into an output of Φ_t . Regard the latter as a finite concatenation of its displayed old words v_j and displayed new separators z . Fix one displayed source separator and take the last displayed target component it meets. A tail of the source separator embeds into this component. Since the separator is indecomposable, the whole separator embeds into that component. The component cannot be an old word v_j . It must be a displayed target separator.

Step 2: the matching of separators is strictly increasing. The tail from Step 1 has length λ , so its image in that target separator is cofinal. No position of a subsequent source separator can follow that cofinal image while remaining in the same target separator. Thus the last target separators of consecutive source separators are strictly increasing. There are t target separators, proving (21).

Step 3: with equal counts, the old coordinates cannot spill. Suppose $m = t$. The increasing assignment between two m -element index sets is the identity. Therefore the i th source separator ends cofinally in the i th target separator, for every i .

Consider an old source word u_i . If $i > 0$, the cofinal image of the preceding source separator prevents every position of u_i from landing in or before the preceding target separator. If $i < m$, all positions of u_i precede the image of the first position of the next source separator. That first image

lies in or before the corresponding next target separator, because that separator is its last target component.

If any position of u_i entered the next target separator, all subsequent positions of u_i would also lie there, strictly below the image of the first position of the next source separator. This would embed a nonempty tail of u_i into a bounded initial interval of a length- λ block. It is impossible by lemma 4.3. Therefore u_i remains wholly inside v_i . The same argument handles u_0 without a left boundary; u_m has no right separator, but the previous separator already forces it into the final target component v_m .

Hence $\Phi_m(\vec{u}) \preceq \Phi_m(\vec{v})$ implies $u_i \preceq v_i$ for all i . Conversely, coordinatewise embeddings together with identity embeddings on z concatenate to an embedding of the outputs. This proves order reflection and preservation.

Finally, finite concatenation of words whose lengths are positive finite multiples of λ , interleaved with words of length λ , again has length a positive finite multiple of λ . $\square$

Remark 6.2 (Why the protection assumption is indispensable). Take the old language to be finite words and $z = a^\omega$. The source coordinate a does not embed into the empty target coordinate, yet $az \equiv z$. Thus merely requiring z not to embed into an old word is insufficient for coordinate reflection. It controls how many separators there can be, but not absorption before a separator. The λ -ended condition is exactly what rules out this failure.

Proposition 6.3 (One additional atom, one additional exponent step). *Under the hypotheses of lemma 6.1, suppose $|B| = M$ and $\text{o}(E) = H_M$. Then $\mathcal{L}(B \cup \{z\})$ contains a λ -ended induced suborder of maximal order type H_{M+1} , and its own maximal order type is H_{M+1} .*

Proof. Take the union of the images of Φ_m for all $m \geq 0$. By lemma 6.1, its layers are copies of E^{m+1} ; an embedding cannot decrease m . Different layers cannot be equivalent, since applying the separator-count inequality in both directions would force their indices equal. Lemma 2.2 therefore gives lower bound H_{M+1} .

Condition (19) ensures that z is not equivalent to a generator in B , so the enlarged generator alphabet has $M + 1$ elements. Apply corollary 3.5 for the matching upper bound. The union is λ -ended by the last assertion of lemma 6.1. $\square$

7 Proof of finite-alphabet exactness

We now assemble the argument for theorem 1.1. The finiteness, canonical order, and recurrence were proved in section 3; only equality in the order-type bound remains.

Proof of theorem 1.1. Induct on k . For $k = 1$, $\mathcal{W}_1(P) = P^*$ and $N_1(P) = n$, so the result is the finite-alphabet Higman formula.

Assume

$$\text{o}(\mathcal{W}_r(P)) = H_{N_r(P)}$$

for some $r \geq 1$, and set $\lambda = \omega^r$ and $N = N_r(P)$. Let Z be the finite poset of the new height- r atoms:

$$Z = \mathcal{I}_{r+1}(P) \setminus \mathcal{I}_r(P).$$

Choose a minimal letter $a \in P$. The constant word a^λ is a minimal element of Z , as shown in section 5. Choose a linear extension

$$z_1, z_2, \dots, z_d$$

of Z that starts with $z_1 = a^\lambda$. Define

$$B_i = \mathcal{I}_r(P) \cup \{z_1, \dots, z_i\} \quad (1 \leq i \leq d).$$

By proposition 5.2, $\mathcal{L}(B_1)$ contains a λ -ended induced suborder E_1 of type H_{N+1} . Suppose, for $i \geq 2$, that $\mathcal{L}(B_{i-1})$ contains a λ -ended induced suborder E_{i-1} of type H_{N+i-1} .

The atom z_i cannot embed into a lower-height atom: its ordinal length is larger. It cannot embed into any z_j with $j < i$ either, because $z_1, \dots, z_d$ is a linear extension of the atom order. Hence $z_i \not\leq b$ for every $b \in B_{i-1}$. Apply proposition 6.3 to obtain a λ -ended induced suborder $E_i \subseteq \mathcal{L}(B_i)$ of type H_{N+i} .

After all d new atoms have been added,

$$B_d = \mathcal{I}_{r+1}(P), \quad N + d = N_{r+1}(P).$$

Its concatenation language is all of $\mathcal{W}_{r+1}(P)$ up to equivalence, by theorem 3.4. The constructed lower bound is $H_{N_{r+1}(P)}$, and corollary 3.5 gives exactly the same upper bound. This completes the induction. $\square$

7.1 The stronger induction invariant

At every positive height the proof establishes slightly more than the statement of the theorem. Once the first atom of that height is activated, each intermediate language already realizes its full Higman upper bound on words ending in an entire block of that height. This invariant is what makes the next separator admissible without adding a new letter to P .

The proof does *not* exhibit an order embedding of the entire free word order $\mathcal{I}_k(P)^*$ into $\mathcal{W}_k(P)$. It exhibits a sequence of induced product layers large enough to attain the same maximal order type. Equality of maximal order types is weaker than an isomorphism of these orders, and absorption shows why that distinction is necessary.

8 The binary cases, worked out explicitly

8.1 Two incomparable letters

Let $P = A_2 = \{a, b\}$. At heights zero and one the atoms are

$$a, \quad b, \quad A = a^\omega, \quad B = b^\omega, \quad C = (ab)^\omega.$$

The nontrivial relations are generated by

$$a < A < C, \quad b < B < C.$$

The two middle atoms A, B are incomparable, as are the two letters. There are exactly five distinct atoms, so the upper bound is H_5 .

The lower-bound construction can be read as three concrete stages. Start with finite binary words, whose type is $H_2 = \omega^\omega$. The map

$$u \mapsto uba^\omega$$

embeds them into words of length ω . Finite products of these blocks, layered by the number of blocks, give a family E_A of ω -ended words with type

$$\sum_{d \geq 1} (\omega^\omega)^{\otimes d} = \omega^{\omega^2} = H_3.$$

Only a, b, A are needed as generators.

Next add B . It embeds into neither a finite letter nor A , so it does not embed into any old word. Interleave m copies of B among $m+1$ independently chosen members of E_A . The protected-separator lemma gives a new ω -ended family $E_{A,B}$ of type

$$\sum_{m \geq 0} H_3^{\otimes(m+1)} = H_4 = \omega^{\omega^3}.$$

Finally add C . It does not embed into any of a, b, A, B : an infinite repetition containing both letters cannot embed into a monochromatic infinite block. The same construction yields

$$\sum_{m \geq 0} H_4^{\otimes(m+1)} = H_5 = \omega^{\omega^4}.$$

This matches the five-generator upper bound and proves the binary antichain instance of corollary 1.2.

Notice that C *dominates* the old infinite atoms. This does not invalidate its use as a separator. What matters is that C does not embed *into* an old atom; protection prevents the old coordinates from being swallowed by a bounded part of a target C .

8.2 Two comparable letters

For $C_2 = \{0 < 1\}$, the height-zero and height-one atoms are

$$0, \quad 1, \quad A = 0^\omega, \quad B = 1^\omega.$$

Here $(01)^\omega \equiv 1^\omega$, because every source letter is at most 1 and the 1 positions occur cofinally. The atom order is a diamond: 0 is below the incomparable atoms 1, A , both of which are below B .

The same activation map $u \mapsto u10^\omega$ gives type H_3 after A is added. Adding B as a protected separator gives H_4 . There are only four atoms, so

$$o(\mathcal{W}_2(C_2)) = \omega^{\omega^3}.$$

Thus two alphabets with the same finite maximal order type 2 already have different exact transfinite-word types.

8.3 All finite alphabets at height two

The height-one atoms correspond to nonempty order ideals of P . An antichain $S \subseteq P$ represents the periodic word $(\prod_{s \in S} s)^\omega$; its class depends precisely on the ideal $\downarrow S$. Therefore

$$N_2(P) = n + J(P) - 1,$$

which proves [corollary 1.2](#) from the main theorem.

Corollary 8.1 (Extremal alphabets of a fixed finite size). *For every n -element poset P , with $n \geq 2$,*

$$\omega^{\omega^{2n-1}} \leq o(\mathcal{W}_2(P)) \leq \omega^{\omega^{n+2^n-2}}. \quad (22)$$

The lower endpoint is attained exactly by chains, and the upper endpoint exactly by antichains.

Proof. Every linear extension of P gives $n + 1$ distinct prefix ideals, so $J(P) \geq n + 1$. If P is not a chain, choose incomparable x, y . The ideals $\downarrow x$ and $\downarrow y$ are incomparable, so at least one is not in the chain of prefix ideals; hence $J(P) > n + 1$. A chain has precisely $n + 1$ ideals.

There are at most 2^n subsets of P . All of them are ideals exactly when P is an antichain: a strict comparison would make the singleton containing its larger element fail to be an ideal. Substitute these bounds into (3). The finite exponent parameter is strictly increasing, so the equality characterizations follow. $\square$

9 Effective enumeration and computed examples

9.1 Canonical generation as a finite algorithm

The recurrence (16) comes with a direct algorithm. Store each leaf by its label and each node by a sorted tuple of child identifiers. At a new height, enumerate the antichains of the already constructed poset; retain those whose largest child height is the preceding height; and create the corresponding nodes. Compute comparisons by (12)–(15), memoizing each ordered pair of identifiers.

This is a decision procedure for all bounded-height atom comparisons. It also gives an exact ordinal notation for the maximal order type: a finite integer $N_k(P)$ together with the fixed expression $\omega^{\omega^{N_k(P)-1}}$. No enumeration of actual infinite words is required.

The finite objects can nevertheless become large. With N existing nodes, there are at most N^2 ordered pairs to memoize and at most N^2 child-pair checks per pair, giving a simple $O(N^4)$ arithmetic-operation bound for the complete comparison matrix. Antichain enumeration can have exponentially many outputs. The code uses integer bit masks and explicit size caps; it makes no claim of a polynomial-time algorithm in k or in the original alphabet size.

9.2 Two independent ideal counters

The package counts $J(R)$ in two ways. Let M be the set of remaining vertices in an induced subposet.

The first method counts antichains as independent sets of the comparability graph. For a chosen $x \in M$, an antichain either omits x or contains x and omits every vertex comparable with x :

$$A(M) = A(M \setminus \{x\}) + A(M \setminus \{y \in M : y \leq x \text{ or } x \leq y\}). \quad (23)$$

The second method directly partitions ideals according to whether they contain x :

$$J(M) = J(M \setminus \uparrow_M x) + J(M \setminus \downarrow_M x). \quad (24)$$

In the first branch, excluding x forces exclusion of everything above it. In the second, the entire downset of x is fixed as present, and one chooses an ideal in the remainder. Both recursions have value one on the empty poset and are memoized. For small posets the code additionally checks all subsets directly against the definition of an ideal.

The two counters share the mathematically specified atom order, but they do not share an antichain-enumeration implementation. Their agreement is therefore a meaningful check of the numerical computations, not an independent proof of the atom-comparison theorem.

9.3 Counts and exact ordinal parameters

Write A_n for the n -element antichain and C_n for the n -element chain. The following are exact values of $N_k(P)$; a dash means not computed in this package.

Alphabet	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
A_2	2	5	11	28	136	—
C_2	2	4	7	12	22	50
A_3	3	10	43	1962	—	—
C_3	3	6	12	28	—	—

Table 1: Canonical indecomposable counts. Each entry N gives $\mathfrak{o}(\mathcal{W}_k(P)) = \omega^{\omega^{N-1}}$.

For example, the theorem and the table give

$$\begin{aligned} \mathfrak{o}(\mathcal{W}_3(A_2)) &= \omega^{\omega^{10}}, & \mathfrak{o}(\mathcal{W}_4(A_2)) &= \omega^{\omega^{27}}, \\ \mathfrak{o}(\mathcal{W}_3(C_2)) &= \omega^{\omega^6}, & \mathfrak{o}(\mathcal{W}_4(A_3)) &= \omega^{\omega^{1961}}. \end{aligned}$$

The last reported value for each family is obtained from the two exact ideal counts at the preceding level. Its full next-level atom poset is not materialized. The CSV records this distinction explicitly.

The first binary counts can be verified without software. The five-element $\mathcal{I}_2(A_2)$ has nine ideals not containing C : choose a prefix in each of the two disjoint two-element chains $a < A$ and $b < B$. There is one more ideal containing C , namely the whole poset. Thus $J(\mathcal{I}_2(A_2)) = 10$ and $N_3(A_2) = 2 + 10 - 1 = 11$.

The six height-two atoms in this case can be represented by

$$\begin{aligned} &\Omega(\{A\}), \quad \Omega(\{B\}), \quad \Omega(\{C\}), \\ &\Omega(\{A, B\}), \quad \Omega(\{A, b\}), \quad \Omega(\{a, B\}). \end{aligned}$$

For the two-element chain, $\mathcal{I}_2(C_2)$ is the four-element diamond and has six ideals, giving $N_3(C_2) = 2 + 6 - 1 = 7$.

10 Growth, interpretation, and limits of the result

10.1 A direct iterated-exponential lower bound

The exact formula also supplies a simple finite-alphabet growth result. For $n \geq 2$, define integers

$$w_0(n) = n, \quad w_{r+1}(n) = \binom{w_r(n)}{\lfloor w_r(n)/2 \rfloor}.$$

Proposition 10.1. *The height- r atoms over A_n contain an antichain of size $w_r(n)$. Consequently, for $k \geq 2$,*

$$N_k(A_n) \geq n + 2^{w_{k-2}(n)} - 1, \quad o(\mathcal{W}_k(A_n)) \geq \omega^{\omega^{n+2^{w_{k-2}(n)}-2}}. \quad (25)$$

Proof. The n leaves give the height-zero antichain. Given an antichain D of height- r atoms, the nodes $\Omega(S)$ for nonempty $S \subseteq D$ are ordered exactly by subset inclusion: no distinct members of D compare. Choosing all subsets of size $\lfloor |D|/2 \rfloor$ produces an antichain of the claimed size at height $r + 1$. For the stated n , this size is positive and the subsets used are nonempty.

Also, all $2^{|D|} - 1$ nonempty subsets give distinct nodes at the next height. Use a height- $(k - 2)$ antichain and retain those nodes together with the n leaves. This proves the first inequality in (25); the second follows from theorem 1.1. $\square$

The elementary bounds

$$\frac{2^m}{m+1} \leq \binom{m}{\lfloor m/2 \rfloor} \leq 2^m$$

follow because the largest binomial coefficient is at least the average of the $m + 1$ coefficients. Thus, for fixed k and growing n , the finite exponent parameter has the expected iterated-exponential growth. The upper estimate

$$N_{k+1}(P) \leq n + 2^{N_k(P)} - 1$$

follows from (16). These are statements about finite alphabets and growth in their size; no sharp asymptotic equivalent is claimed.

10.2 The quotient, not the literal set of words

There may be uncountably many literal words of length ω over a finite alphabet, but at each fixed finite level there are only countably many equivalence classes of all words. Indeed, every class has a finite code over the finite atom alphabet. The large ordinals in the theorem measure ordering complexity of this quotient, not the cardinality of the original set of functions.

The integer $N_k(P)$ should also not be mistaken for the number of all word classes. It counts only indecomposable classes. Their finite concatenations have infinitely many classes and supply the much larger maximal order type.

10.3 Why this does not compute the all-height union

Although

$$\mathcal{W}_{<\omega^\omega}(P) := \{u : |u| < \omega^\omega\} = \bigcup_{k \geq 1} \mathcal{W}_k(P),$$

one cannot infer its maximal order type merely by taking the supremum of the numbers in [theorem 1.1](#). Maximal order type is not generally continuous under increasing unions of induced suborders.

A simple example is the increasing union of the finite squares $\{0, \dots, m-1\}^2$ under component-wise order. Each has maximal order type m^2 , so the supremum of their types is ω . The union is $\omega \times \omega$, whose maximal order type is $\omega \otimes \omega = \omega^2$. The all-height problem requires its own argument; the broader tree correspondence and all-height results in [2] are separate from the finite-level proof here.

10.4 What remains outside the theorem

The proof uses finiteness in two different ways. The normal-form step needs only finitely many lower-height atom classes, so that recurrent types can be isolated after a finite prefix. The exactness step adds all new atoms in a finite linear extension and uses the finite-letter Higman formula. Neither argument directly provides an exact answer for every infinite wqo alphabet, arbitrary transfinite height, or arbitrary cutoff ordinal that is not a power ω^k .

Nor does a large maximal order type automatically specify an optimal length-function bound for finite controlled bad sequences. Such a bound also requires a norm and a control function. The ordinal theorem is an input to that separate analysis, not a substitute for it.

11 Verification, proof audit, and reproducibility

11.1 What was actually checked

The code was run successfully under Python 3.13.5, using only the standard library. It is written to support Python 3.9 or later. The fixed random seed is 20260919. The recorded run reports:

Check	Count
All labelled posets on at most three elements	24
Canonical atom posets through $k = 3$ for those alphabets	72
Agreements of two independent ideal counters	87
Agreements with direct exhaustive subset counting	64
Marker-map comparison pairs	1,922
Equal-block comparison pairs	10,000
Separator comparison pairs	18,000
of which equal-separator coordinate checks	13,489
Singleton length-order comparison pairs	625
Absorption sanity checks	2
Exact count-table entries	19

Table 2: The completed reproducibility run. These are finite checks, not a formal verification of the transfinite proof.

The count “24” includes one empty poset, one one-element poset, three labelled two-element

posets, and nineteen labelled three-element posets. The atom checks verify reflexivity, antisymmetry, transitivity, height monotonicity, and antichain reduction. The symbolic word checks operate on exact ultimately periodic ω -blocks; they do not replace ω by a large finite integer. Section B describes the embedding algorithm and its correctness argument.

These checks exercise the finite representations, detect arithmetic and implementation mistakes, and test many instances of the structural lemmas. They do not establish arbitrary-height statements by exhaustive search. The latter are supported by the written induction and cofinality proofs.

11.2 The main points audited in the proof

The first possible failure is assuming that a surjective concatenation map is order-reflecting. It need not be; the proof uses it only for the upper bound. Order reflection for the lower bound is proved separately on the protected families.

The second is confusing “has the same cardinality” with “has the same ordinal length.” All no-spill arguments compare ordinal order types. A tail of a λ -ended word has length at least λ , and cannot fit below a single position of a target λ -block.

The third is silently treating an indecomposable as wholly contained in its last target factor. The original embedding may straddle factors. Lemma 3.2 extracts an equivalent tail to obtain a new embedding; lemma 6.1 then uses cofinality of that tail in its last target separator.

The fourth is choosing separators in the wrong order. The requirement is that the *new* atom not embed into any *old* atom. An increasing linear extension of the same-height atom poset supplies exactly that direction. Old atoms are allowed to embed into the new one.

The fifth is combining product layers without checking cross-layer comparisons. In activation, length makes the layer index monotone. In separator amplification, equation (21) does so. This also precludes identifying outputs from two different layers on the quotient.

Finally, the singleton alphabet is checked separately. It has one atom at each height, so $N_k(1) = k$, but its word type is ω^k , not H_k for $k \geq 2$. The missing second letter is a genuine mathematical obstruction to the activation argument, not a cosmetic exception.

11.3 How to reproduce the package

From the archive root, run

```
python code/verify.py --out data
```

The program regenerates data/verification.json, the CSV count table, and explicit JSON atom posets. To inspect a particular alphabet, for example the binary antichain through height two, run

```
python code/atoms.py --kind antichain --size 2 --k 3 --output atoms.json
```

A custom finite poset may be supplied as JSON containing distinct labels and a full reflexive Boolean relation matrix. The default cap prevents materialization of more than 5,000 atoms; the cap can be changed explicitly. Counting can still be expensive before a large object is materialized.

Build this document using

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is included in the source, so no bibliography database or network access is needed. A separate `references.bib` is provided for reuse. The original third-party papers and font files are not included.

A The finite-alphabet Higman formula

This appendix records a proof of the finite case of (7), using Higman’s wqo theorem and the standard calculus (5). It also explains why the separator construction in the main argument has the right ordinal shape.

Theorem A.1 (Classical finite-alphabet formula). *For a nonempty finite poset R with n elements,*

$$o(R^*) = \omega^{\omega^{n-1}} = H_n.$$

Proof. Induct on n . For $n = 1$, words are ordered by their finite lengths, so the type is $\omega = H_1$. Assume $n \geq 2$ and the result for smaller alphabets.

Lower bound. Choose a maximal letter $b \in R$ and set $Q = R \setminus \{b\}$. In words

$$u_0 b u_1 b \cdots b u_m, \quad u_i \in Q^*,$$

every source b must map to a target b , since $b \not\leq q$ for $q \in Q$. The number of displayed b ’s cannot decrease under embedding. When the two words have the same number, all displayed b ’s match in order, forcing the intervening coordinates into their corresponding intervals. Consequently, the layer with m occurrences of b is an induced copy of $(Q^*)^{m+1}$. The layers are disjoint and their index is monotone, so lemma 2.2, with $o(Q^*) = H_{n-1}$, gives lower bound H_n .

Upper bound. Fix a forbidden nonempty word $w = p_1 \cdots p_\ell$. A word that does not contain w as an embedded subsequence has a greedy factorization

$$u_1 b_1 u_2 b_2 \cdots u_j b_j u_{j+1}, \quad 0 \leq j < \ell,$$

where $b_i \geq p_i$ and u_i uses only letters in $Q_i = R \setminus \uparrow p_i$. Each Q_i is a proper subalphabet. This factorization is obtained by greedily matching p_1 , then p_2 , and so on, stopping when the next letter cannot be matched.

For a fixed j and fixed matched letters b_i , concatenation is a monotone surjection from a product of $j + 1$ orders Q_i^* onto this part of the residual. By induction every nonempty Q_i^* has type at most H_{n-1} ; an empty Q_i contributes the one-element order, which also satisfies the bound. There are only finitely many choices of j and of the b_i . Thus the residual $R^* \setminus \uparrow w$ is a monotone image of a finite disjoint union of products, whose maximal order type is bounded by a finite natural sum of ordinals at most

$$H_{n-1}^{\otimes \ell} = \omega^{\omega^{n-2} \cdot \ell}.$$

Every such finite sum is strictly less than H_n . The residual for the empty forbidden word is empty. Using the residual identity in (5) and the fact that H_n is a limit ordinal gives $o(R^*) \leq H_n$. $\square$

The transfinite proof differs in one crucial respect. A new separator is an infinite word, not a single letter. Old coordinates can in principle enter its beginning even when its occurrences have all been matched. Protected tails are what restore the coordinate conclusion.

B Exact symbolic embedding below ω^2

The test program represents a word as

$$p_1 c_1^\omega p_2 c_2^\omega \cdots p_t c_t^\omega q, \quad (26)$$

where each p_i, q is a finite word and each c_i is a finite nonempty cycle. This is an exact symbolic representation of a transfinite word, not a truncation. It is sufficiently general to represent every class below ω^2 over a finite alphabet, by the normal-form argument, although the tests use bounded finite parameter sets.

B.1 The greedy algorithm

Maintain a cursor in the target. Within a target block it consists of its block index and a finite offset in its finite prefix or periodic tail. After the final infinite block it is an offset in the finite suffix. Process a source finite prefix one letter at a time, always choosing the earliest target position carrying a label above the source label. In a periodic tail, checking one cycle suffices to find the earliest match; if no cycle letter is suitable, skip to the next target block.

For a source periodic tail c^ω , find the earliest target infinite block, at or after the cursor, whose recurring cycle d satisfies

$$(\forall a \text{ occurring in } c)(\exists b \text{ occurring in } d) a \leq_P b. \quad (27)$$

If there is no such block, embedding fails. Otherwise embed the periodic tail cofinally into that block and advance the cursor to the beginning of the next target block. Continue with the next source prefix. After all source infinite blocks are consumed, greedily process the finite source suffix.

B.2 Why the algorithm is exact

Greedy matching of a finite letter is safe because replacing its target position by an earlier suitable one can only enlarge the remaining target interval. This is the usual finite-subsequence argument and applies unchanged inside a transfinite target.

An infinite periodic source tail has infinitely many occurrences of every label in its cycle. If it embeds into a finite concatenation of target infinite blocks and a finite suffix, its image ends cofinally in one target infinite block. A source tail of the same periodic type therefore embeds into that block's recurring tail. This forces (27); finite exceptional target prefixes cannot supply infinitely many occurrences of a missing label.

Conversely, (27) suffices: place successive source letters in suitable positions of successive sufficiently late target cycles. Any finite cursor offset is harmless. Choosing the earliest compatible target block leaves the largest possible collection of whole target blocks for subsequent source material. No subsequent source position could remain in the same target block after a cofinal image. Thus advancing to the next block is both sound and optimal. Induction over the finitely many source blocks proves correctness.

B.3 Test families and negative controls

The marker tests compare all finite binary source words of length at most four with all finite binary target words of length at most four, for both the chain and antichain letter orders. There are $2(1 + 2 + 4 + 8 + 16)^2 = 1,922$ pairs.

The equal-block tests use up to four exact ω -blocks, with finite prefixes and one of the cycles 0, 1, or 01. The separator tests first add 1^ω to a language whose infinite blocks have cycle 0. For the antichain they also add $(01)^\omega$ to a language whose infinite blocks are monochromatic. Both equal and unequal separator counts are tested, including positive comparisons and self-comparisons.

As negative controls against an incorrectly rigid concatenation implementation, the program verifies the genuine absorption $aa^\omega \equiv a^\omega$. It also checks that over a singleton alphabet, the embedding decision agrees exactly with comparison of the ordinal lengths $\omega t + s$ on the finite test range.

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